

POLS 641

Introductory Political Analysis

Class 1: Introduction

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Today's Agenda

- Research in Action- Latte Liberals
- Introductions
- Introducing Quantitative Methods
- Introducing the Syllabus

- Thursday: Introducing R

Latte Liberals

- What is the research question?
- What is the dependent, or outcome, variable?
- What is the independent, or explanatory, variable?
- Methods Terms: Spurious, Causal, Correlation, Hypothesis, Representative Sample, index, Probit Regression, dichotomous outcome, control variable, coefficient, ($p < 0.05$)

Figure 1

Four Possible Explanations for a Spurious Relationship between Liberal Ideology and Latte Drinking

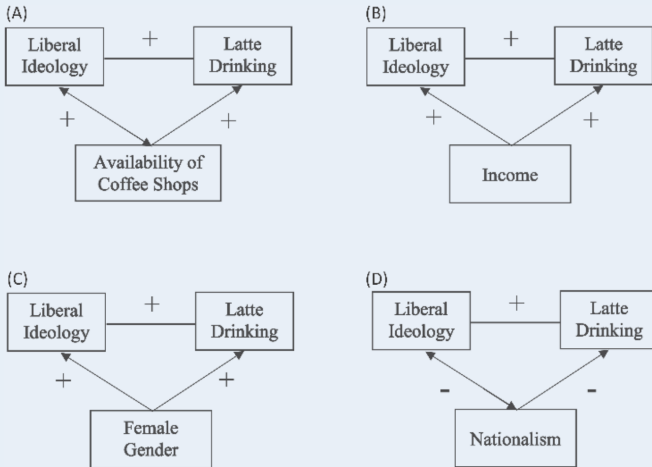


Table 1

Predictors of Latte Drinking, Controlling for Potential Explanations

	Initial Model	Model A	Model B	Model C	Model D	Saturated Model
Liberalness	0.08 (0.03)*	0.08 (0.03)*	0.08 (0.06)*	0.08 (0.03)*	0.01 (0.04)	0.01 (0.04)
Availability of Coffee Shops		0.09 (0.03)*				0.10 (0.04)**
Income			0.05 (0.01)***			0.06 (0.01)***
Female Gender				0.26 (0.10)*		0.31 (0.12)**
Nationalism					-0.38 (0.09)***	-0.35 (0.09)***
Cutpoint	1.52 (0.14)	1.59 (0.15)	2.12 (0.21)	0.54 (0.26)	0.14 (0.33)	1.16 (0.40)
Sample Size	1,059	1,059	1,059	1,059	896	896

Notes: Entries are probit coefficients with standard errors in parentheses. Results are based on probit regressions with the dependent variable coded as Prefer Lattes (1) or Not (0). Liberalness is coded on a 7-point scale. Analysis based on the subsample of self-identified coffee drinkers.

Introductions

- Name
- Subfield
- Where From/Degree(s)
- Current Research Interests
- Background in Methods and Stats
- What do you do for fun/health

Introductions

About Me:

- PhD Political Science - University of California, Santa Barbara
- Research Interests: Environmental Politics, Global Governance, Pro-Environmental Behavior, Sustainability
- Methods I've Used in My Research: Time Series Regression, Dyadic Regression, Surveys, Survey Experiments, Field Experiments, Factor Analysis, Process Tracing, Item Response Theory, Meta-Analysis, Psychometrics, Text-as-data

Why Research?

Social "Science"?!? My take...

- A search for truth
- An attempt to describe and explain social phenomena (Humans are complicated)
- A relatively standard operating procedures that help us do do in a productive and cumulative way ("scientific method")
- a social purpose: a hope that increased understanding can lead to some type of social improvement

Political Science

- Politics uses the methods of rhetoric and advocacy to persuade people to adopt a certain point of view.
- Political science uses the scientific method to uncover the true state of the world

Broad Categories of Political Science Methods

Interpretation

Qualitative/Process Tracing

Simulation

Experiments

Large-N

Experiments

Ideal: Fully Randomized Control Trial

Good For: Causation

Bad For: Generalizability & External Validity

Large-N

Look for patterns in a large number of cases
. . . . usually using a form of regression

Causation is key

$$X \rightarrow Y$$

Our Goal is Inference

- Inference is the process of using facts we do know to learn facts we didn't know before.
- Descriptive inference is using these facts to learn something about the state of the world.
- Causal inference is using these facts to learn something about how the world works.
- Our focus will be on **causal inference**.

The Goal of (Quantitative) Research

- Underlying everything we're doing is the concept of *causality*.
- Deterministic versus probabilistic causality.
- Most of our hypotheses will focus on approaching causality.
- “All else equal (*ceteris paribus*), an increase in X causes an increase in Y.”
- In this course, we will learn tools to set up and test causal hypotheses
- Establishing causality often comes at the cost of generalizability

This Course

A graduate-level introduction to regression analysis

The course covers:

- Part I: Basic principles of probability and statistics Essential for understanding how regression works
- Part II: Linear regression analysis The workhorse method in quantitative social science

After this course, you will be:

- Capable of conducting basic data analysis
- Able to critically read and understand quantitative empirical studies published in academic books and journals
- Be prepared to learn more advanced statistical methods



Questions?

Syllabus

Course Structure

Our *often overlapping* tasks

- 1 Understand the core conceptual and theoretical foundations of research design, probability, statistics, and data analysis
- 2 Application using R including proper coding, data management, visualization, and interpretation.

Weekly Schedule

- Tuesday - Course concepts discussion and lecture
- Wednesday midnight - Datacamp/Problem Set due
- Thursdays - R work
 - Office Hours Prof Kuehl: Tuesday 10- 11 and Thursday 330-430 & by appointment

Course Requirements

Weekly Schedule

- Final Project (50%)
- Problem Sets (25%)
- Midterm (10%)
- Participation (10%)
- Datacamp (5%)

Course Policies

- Group work - Encouraged, but do your own
- R Guru
- Email and Office Hours

Course Policies- AI and LLMs

You MAY use AI to:

- find outside sources
- troubleshoot technical issues
- sparingly to debug error and warning messages

You MAY NOT use AI to:

- Think for you
- Do any writing for this course(including problem sets)
- summarize course readings
- assist during midterm

The struggle causes the learning

Graduate School Grades

- The course is not about grades (it's about learning)
- **A** : *Exceptional Performance*. Outstanding work on all course-related tasks at a level that distinguishes the student from other members of the class. A comprehensive and incisive command of the issues, literature, and substantive information relevant to the course. The ability to master and integrate large amounts of factual material and abstract theories.
- **A-** : *Very Good Performance*. Consistently strong work on all course-related tasks. A command of the issues, literature, and substantive information relevant to the course. Understands well and can integrate the relevant factual and theoretical material central to the course.
- **B+** : *Good Performance*. Satisfactory work on all course-related tasks. A good grasp of the issues, literature, and substantive information relevant to the course. An OK command of factual and theoretical material, and some capacity to integrate the two.

Advice from your peers 1

What advice would you give your week 1 self?

- It is easy to fall behind on the problem sets and get stuck doing it last minute, especially when the stress of the semester hits. Its easier said than done, but try to start working on the Psets as soon as it is released. It is better to tackle a few questions here and there and then put it away, than leave ave it all till few days before it is due to try to do it.
- I would tell myself to take my time, don't let datacamp upset me too much, work with others, and do the readings. I did all these things (except for not getting upset with data camp when it started to be ridiculous), but they really help. The readings help phrase things in different ways, and helped me come to class with questions, which were often answered by your presentations. Working on the problem sets with other people was great for when I was totally over thinking things, as is easy to do when staring at data for a while. It was like fresh eyes without actually having to take more time.

Advice from your peers 2

- Ask more, you will get overwhelmed with homework and assignment. Make friends with your cohort in the class they will be your assets if you need help. Professors are busy people, Colin is good instructor, but for someone never took any coding classes, he was talking Chinese during the Tuesdays classes. I'm grateful I can code and analysis data now, it took a lot to get to this point.
- The first thing I would suggest to my week 1 self is to trust the professor that by the end of the sixteen weeks you will have a grasp on the coding and statistics that sound SO scary to you on that first day. I would also suggest talking through issues you may have with your classmates, or answering any questions that they may have, because I found that 1. it can help clarify any problems (googling only helps to a degree) and 2. it also helps you reinforce your knowledge by answering other people's questions.

Advice from your peers 3

- Worry less about the coding and more about interpreting the results. I think the coding seems like a big hurdle to overcome since it is completely new unless someone has previous experience with computer languages but the more important aspect is to be able to interpret the output of the code. The coding becomes easier the more you do it, as does interpretation, but a new student should worry more about the interpretation than getting the code correct.
- First, even when it feels like everyone knows everyone and you are lost, not everyone knows everything about coding they are learning it new just like you. Two, ask more questions and do not worry about looking stupid because once you gloss over the first thing you do not understand it gets worse. Third, breathe coding is hard but understanding it makes you feel like a superhero.

Advice from your peers 4

- Don't cry, it isn't worth it. Find some buddies and work through the problem sets together. You'll retain the information better and get work done quicker as it keeps you accountable.
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Advice from your peers 2025

- 1. Just trust Prof Kuehl and your R-Guru/TA.
- 2. Stop pushing yourself too much and losing the weekends just to get your stuff done.
- 3. Give yourself a day or two with complete detachment from assignment (you will still have time for it!).
- 4. Find a friend that you can ask to hang out.
- 5. Get out of your room, you only make it worse.
- 6. Eat healthier and exercise daily. It will help you to maintain a good physical rhythm and provide you with a resilient physical capacity to face any stress and surging cortisol level throughout your first ever semester in NIU.
- 7. Failure is a part of learning, embrace it, look for advice and constructive feedback, and do it all over again.
- 8. Know that our campus has all valuable resources for academic coaching, research assisting, mental health and counselling, and the most important one is the clinic/health center.

Advice from your peers 2025

Letter from the past (to next year's students): For those who are afraid with number or statistics, worry not, my friend. I am historian, never had statistics class since undergraduate. However, curiosity will push you even further. Who knows that now, a historian, can read and understand the number and Greek symbol in regression table? Do your problem sets and Datacamp as best as you can. In my case, it helps me to keep building. Those are like step ladder or scaffolding which help you understand the idea behind the music

The Grad School Toolbelt - A Unsolicited Note on Software

- Automatic backup!!!! (well organized)
- Bibliographic management(Organization, notes, citations)
- Statistical Analysis
 - transparent
 - reproducible
 - exportable
- Editor that can:
 - Weave it all together
 - Let you write!
 - Create professional output

Software Options

- Backup: OneDrive, Dropbox, Google Drive, Github
- Bibliographies: Zotero, Mendeley, JabRef, Papers
- Statistical Analysis: R with RStudio I
- Editor: MS Word, RMarkdown in RStudio, Google Docs, $\text{\LaTeX}$

Grad School Advice

- Focus on the big picture(grades don't matter)
- Your time has suddenly become much more valuable
- You will learn more from your peers than you will from your professors
- Read Smart
- It is a marathon, not a sprint
- You will get out of this what you put into it

Reading for Next Week

Bailey Chapter 1

Angrist and Pischke. 2009. "Questions about Questions" Bush and

Clayton. 2022. "Facing Change: Gender and Climate Change Attitudes Worldwide" *American Political Science Review*

- What is the goal of this research?
- What methods are used to address their research question?
- What are they explaining?
- What is doing the explaining?
- What are some methodological holes in their argument?

What is a scientist after all? It is a curious man looking through a keyhole, the keyhole of nature, trying to know what is going on.

-JACQUES COUSTEAU